Learning Contact rich Skills from Behaviour Priors: Variability-Driven Variable Impedance Control with State-Driven Task Parametrization, Stability and Passivity Guarantees

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Abstract

This report presents a simulated experimentation framework for testing the Thesis' approach for transferring a contact rich skill to a robot manipulator by learning from encoded human behaviour priors. The experimental setup represents a robot with a scalpel attached to its end effector learned to perform an adaptive cutting skill on variant types of materials. unlike replaying a single trajectory, the method learns the pattern of the skill and reproduces it under a variable-impedance controller with an implemented energy-tank passivity mechanism at the joint torque port. The contribution comprises three elements. First, a *variable impedance controller* whose stiffness matrix is learned from the consistency of the demonstrations: a quadratic program (QP) solved at each control step maps the demonstration precision onto the controller gains, so the manipulator is stiff only along directions that the demonstrations regulated and compliant elsewhere. Second, a *state-driven task parametrisation* that indexes the motion by a progress (phase) variable whose evolution is corrected by the manipulator's measured state, providing temporal-scale adaptability and disturbance rejection. Third, a formal *stability and passivity* analysis by projecting the torque command onto a runtime energy-tank power budget, and a positive power perturbation validation demonstrates that the tank prevents energy injection that the controller might generate without tank projection for maintaining robot passivity and safety-aware interaction with human. Exponential convergence is derived under the stated regulation assumptions and is reported empirically as a diagnostic on the regulated task error. An important consideration for the studied approach is the data repeatability to decide which skill patterns to extract under which constraints, thus, across repeated demonstrations the scalpel *position* is highly repeatable (cross-trial correlation ≈ 0.95), whereas the *force* is not (≈ 0.10). Therefore, The controller regulates position and treats desired force as a constraint, while a separate per-material *cutting-force model*, identified from each material's own data, provides the interaction force in simulation. The experiment framework is evaluated on three materials (cork, PVC, penoplex) for both a fixed and a moving workpiece scenarios.

Contents

Nomenclature

The table below lists every symbol used, its meaning, and its physical dimension. Bold lower-case symbols are 3-vectors expressed in the world frame; n is the number of robot joints ($n = 7$ for the KUKA iiwa 14 used here).

Symbol	Meaning	Units
ϕ	task phase (progress along the skill), 0 at start, 1 at end	dimensionless
$x,\ x_d$	actual / desired scalpel-tip (TCP) position	m
$\dot{x},\ \dot{x}_d$	actual / desired scalpel-tip velocity	$\mathrm{m\,s^{-1}}$
$\tilde{x} = x_d - x$	position tracking error	m
$\dot{\tilde{x}} = \dot{x}_d - \dot{x}$	velocity tracking error	$\mathrm{m\,s^{-1}}$
f	commanded Cartesian force (impedance "virtual force")	N
f_{ext}	external (environment/contact) force on the scalpel	N
$K(\phi)$	diagonal stiffness matrix (per axis)	$\mathrm{N\,m^{-1}}$
$D(\phi)$	diagonal damping matrix	$\mathrm{N\,s\,m^{-1}}$
K_I	integral gain	$\mathrm{N\,m^{-1}s^{-1}}$
ζ	damping ratio (set to 0.7)	dimensionless
$\Lambda,\ M$	task-space (Cartesian) inertia	kg
$C\dot{q},\ g$	Coriolis/centrifugal and gravity joint torques	N m
τ	commanded joint torques	N m
$q, \dot{q}$	joint angles / velocities	rad, $\mathrm{rad\,s^{-1}}$
J	TCP Jacobian ($\dot{x} = J\dot{q}$), $3 \times n$	$\mathrm{m\,rad^{-1}}$
λ^2	damped-least-squares regulariser	$\mathrm{m^2\,rad^{-2}}$
N	null-space projector ($I - J^+J$)	dimensionless
$N_{\mathrm{seed}}, N_{\mathrm{demo},*}, N_{\mathrm{mat}}$	numbers of random seeds, demonstrations, and material-identification samples	dimensionless
$\sigma_i^2(\phi)$	cross-trial variance of (range-normalised) signal, axis i	dimensionless
$\Gamma_\chi, \mathcal{R}_\chi$	shape-correlation and regulation scores for signal channel χ	dimensionless
B_χ, U_χ	between-phase and within-phase variances of channel χ	channel units squared

Symbol	Meaning	Units
N_ϕ, N_{boot}	phase samples per trial and bootstrap resamples	dimensionless
$\epsilon_\sigma, \epsilon_\tau$	precision and torque-projection numerical regularisers	dimensionless, $\text{rad}^2\,\text{s}^{-2}$
$w_i(\phi)$	precision weight (normalised inverse variance) $\in [0,1]$	dimensionless
$K_i^{\text{pos}}(\phi)$	precision-derived position stiffness, axis i	$\text{N}\,\text{m}^{-1}$
$K_{\text{floor}}, K_{\text{min}}, K_{\text{max}}$	stiffness floor / box limits	$\text{N}\,\text{m}^{-1}$
u, u_w	normalised candidate / identified stiffness allocations	dimensionless
$H(\phi)$	phase-dependent task-relevance matrix in the allocation proof	dimensionless
$q_f(\phi)$	force-consistency weight in the QP $\in [0,1]$	dimensionless
F_d	desired interaction force (from the prior)	N
F_i^{bound}	per-axis demonstrated force limit	N
$\hat{\phi}$	state-estimated phase	dimensionless
$\dot{\phi}_{\text{nom}}$	nominal (clock) phase rate	s^{-1}
k_ϕ	phase-correction gain $\in [0,1]$	dimensionless
ρ_x, ρ_F	normalising scales for position / force matching	m, N
ℓ_ϕ	forward phase-search horizon	dimensionless
$\mathcal{S}$	total passivity storage (mechanical energy plus tank)	J
V	stability Lyapunov function	J
T	energy-tank state	J
h_c, h_{sim}	controller and simulator integration periods	s
P_{inj}	power drawn by time-varying gains / moving reference	W
β	Lyapunov cross-term coefficient	s^{-1}
α	nominal exponential decay rate	s^{-1}
A	nominal stability state matrix	mixed state-space units
P, Q	Lyapunov and dissipation matrices in the stability proof	mixed energy units
d	blade penetration depth below the material surface	m
k_{mat}	material loading stiffness	$\text{N}\,\text{m}^{-1}$
$F_{\text{mat}}^{\text{cut}}$	material cutting-force plateau	N

1 Problem statement

The robot's task is to acquire a contact-rich manipulation skill, cutting, from a handful of human encoded behaviour priors. The behaviour priors are provided to the robot controller from behaviour prior probabilistic generator trained to encode human demonstrated position, twist and wrench profiles as input to the robot control algorithm. Each demonstration provides, sampled over time, the scalpel pose, its twist (linear/angular velocity), and the interaction wrench measured at the scalpel. This aims to capture the *skill* instead of cloning the behaviour. i.e., a controller that performs the task accurately, respects the force character of the demonstrations, stays stable and safe in contact, and generalises across execution speeds and across materials.

The position and force dependency during contact through the motion path represents the core challenge in the study. In a direction where the scalpel presses on the material, fixing the position determines the force and vice-versa; commanding robot controller to track both is non trivial issue. A principled controller must therefore *decide what to regulate in each direction*, and a reliable simulation must *model the material* that gives rise to the contact force in the first place. Both points are addressed.

2 Patterns in the demonstration that represent a reliable skill

The reliable skill patterns were extracted from data repeatability across trails. i.e, the repeatable components of the demonstration to be treated as the skill where other components are constraints to shape the behaviour of performing the skill by robot. For each signal, across trials measurements of (a) the *shape correlation* and (b) a *regulation score* were reported. shape correlation measures how similarly the signal evolves from trial to trial after removing its mean where *regulation score* refers to the fraction of the signal's variance that is explained by the task phase, i.e. a phase-locked signal-to-noise ratio.

Let $y_{\nu,\chi}(\phi_\ell)$ denote channel χ in demonstration $\nu \in \{1,\ldots,N_{\text{demo}}\}$ after all trials have been resampled at the common phase samples ϕ_ℓ, $\ell \in \{1,\ldots,N_\phi\}$, with $N_\phi = 300$. The trial-specific temporal mean and demeaned profile are

$$\bar{y}^{\phi}_{\nu,\chi} = \frac{1}{N_\phi}\sum_{\ell=1}^{N_\phi} y_{\nu,\chi}(\phi_\ell), \qquad \breve{y}_{\nu,\chi}(\phi_\ell) = y_{\nu,\chi}(\phi_\ell) - \bar{y}^{\phi}_{\nu,\chi}. \tag{1}$$

Here ν indexes demonstrations, χ identifies one of X, Y, Z, F_x, F_y, F_z, and ℓ indexes phase. Subtracting $\bar{y}^{\phi}_{\nu,\chi}$ removes constant trial-dependent offsets before shape comparison. The pairwise shape correlation is

$$\mathrm{corr}^{(\chi)}_{\nu\mu} = \frac{\displaystyle\sum_{\ell=1}^{N_\phi} \breve{y}_{\nu,\chi}(\phi_\ell)\breve{y}_{\mu,\chi}(\phi_\ell)}{\sqrt{\displaystyle\sum_{\ell=1}^{N_\phi} \breve{y}^2_{\nu,\chi}(\phi_\ell)}\sqrt{\displaystyle\sum_{\ell=1}^{N_\phi} \breve{y}^2_{\mu,\chi}(\phi_\ell)}}, \qquad \nu < \mu, \tag{2}$$

and the reported channel statistic is the mean over all distinct trial pairs,

$$\Gamma_\chi = \frac{2}{N_{\text{demo}}(N_{\text{demo}}-1)}\sum_{\nu<\mu} \mathrm{corr}^{(\chi)}_{\nu\mu}\,. \tag{3}$$

Thus $\Gamma_\chi \in [-1, 1]$ measures reproducibility of the evolving profile, not agreement of its absolute offset.

The phase-conditioned mean and grand mean are

$$\bar{\breve{y}}_\chi(\phi_\ell) = \frac{1}{N_{\text{demo}}}\sum_{\nu=1}^{N_{\text{demo}}} \breve{y}_{\nu,\chi}(\phi_\ell), \qquad \bar{\breve{y}}^{\text{all}}_\chi = \frac{1}{N_\phi}\sum_{\ell=1}^{N_\phi} \bar{\breve{y}}_\chi(\phi_\ell). \tag{4}$$

The between-phase and within-phase variances are then

$$B_\chi = \frac{1}{N_\phi}\sum_{\ell=1}^{N_\phi}\left(\bar{\breve{y}}_\chi(\phi_\ell) - \bar{\breve{y}}^{\text{all}}_\chi\right)^2, \tag{5}$$

$$U_\chi = \frac{1}{N_{\text{demo}}N_\phi}\sum_{\nu=1}^{N_{\text{demo}}}\sum_{\ell=1}^{N_\phi}\left(\breve{y}_{\nu,\chi}(\phi_\ell) - \bar{\breve{y}}_\chi(\phi_\ell)\right)^2. \tag{6}$$

B_χ quantifies the common trajectory variation explained by phase, while U_χ quantifies trial-to-trial variation remaining at the same phase. The regulation score is

$$\mathcal{R}_\chi = \frac{B_\chi}{B_\chi + U_\chi} \in [0,1]. \tag{7}$$

A value near one indicates that the signal is predominantly phase locked; a value near zero indicates that knowing the phase does not predict the measured channel reliably across demonstrations.

Proposition 1 (Reliability interpretation of the two metrics)**.** *Suppose an aligned channel admits the repeated-trial decomposition*

$$y_{\nu,\chi}(\phi) = o_{\nu,\chi} + \psi_\chi(\phi) + \upsilon_{\nu,\chi}(\phi), \tag{8}$$

where $o_{\nu,\chi}$ is a trial-dependent constant offset, $\psi_\chi(\phi)$ is a common phase-locked pattern, and $\upsilon_{\nu,\chi}$ is zero-mean trial-specific variation independent across trials and uncorrelated with ψ_χ. If

$$\mathrm{Var}_\phi[\psi_\chi] = \varsigma^2_{\psi,\chi}, \qquad \mathrm{Var}_\phi[\upsilon_{\nu,\chi}] = \varsigma^2_{\upsilon,\chi}$$

is the same for all trials, then, in the large-sample limit,

$$\mathbb{E}\left[\mathrm{corr}^{(\chi)}_{\nu\mu}\right] = \frac{\varsigma^2_{\psi,\chi}}{\varsigma^2_{\psi,\chi} + \varsigma^2_{\upsilon,\chi}}, \qquad \mathcal{R}_\chi \longrightarrow \frac{\varsigma^2_{\psi,\chi}}{\varsigma^2_{\psi,\chi} + \varsigma^2_{\upsilon,\chi}}. \tag{9}$$

Consequently, both statistics estimate the phase-locked fraction of the dynamic signal and are invariant to the constant offset $o_{\nu,\chi}$.

Proof. Demeaning (8) by (1) eliminates $o_{\nu,\chi}$. For distinct trials $\nu \neq \mu$, independence of the residuals and their lack of correlation with ψ_χ give

$$\mathrm{Cov}_\phi(\breve{y}_{\nu,\chi}, \breve{y}_{\mu,\chi}) = \varsigma^2_{\psi,\chi},$$

whereas each demeaned trial has variance $\varsigma^2_{\psi,\chi} + \varsigma^2_{\upsilon,\chi}$. Division of the covariance by the two standard deviations yields the first expression in (9). For the variance decomposition, the phase-conditioned demeaned trial mean converges to the centred common pattern $\psi_\chi(\phi) - \mathbb{E}_\phi[\psi_\chi]$, so $B_\chi \to \varsigma^2_{\psi,\chi}$, while deviations about that mean give $U_\chi \to \varsigma^2_{\upsilon,\chi}$. Substitution into (7) yields the second expression. Neither limit depends on $o_{\nu,\chi}$. □

Quantity	cross-trial shape correlation	regulation score
Position (X/Y/Z)	0.92–0.98	up to 0.73 (Z)
Force (Fx/Fy/Fz)	0.04–0.27	≈ 0

Table 2: How repeatable each quantity is across demonstrations (cork $N_{\text{demo,cork}}$=28, PVC $N_{\text{demo,PVC}}$=33, penoplex $N_{\text{demo,peno}}$=22, 300 samples per trial).

The numerical separation between the two signal classes is

$$m_\Gamma = \min_{\chi \in \{X,Y,Z\}} \Gamma_\chi - \max_{\chi \in \{F_x,F_y,F_z\}} \Gamma_\chi \geq 0.92 - 0.27 = 0.65 > 0. \tag{10}$$

Thus even the least repeatable position channel has substantially greater shape agreement than the most repeatable force channel. Under Proposition 1, either reliability statistic $q_\chi \in \{\Gamma_\chi, \mathcal{R}_\chi\}$ corresponds to the phase-locked signal-to-residual ratio

$$\mathrm{SNR}_\chi = \frac{\varsigma^2_{\psi,\chi}}{\varsigma^2_{\upsilon,\chi}} = \frac{q_\chi}{1 - q_\chi}. \tag{11}$$

The position-correlation range 0.92–0.98 therefore corresponds to an estimated phase-locked ratio of 11.5–49, whereas the force-correlation range 0.04–0.27 corresponds to only 0.04–0.37. Independently, the depth regulation score $\mathcal{R}_Z = 0.73$ gives $\mathrm{SNR}_Z \approx 2.70$, while $\mathcal{R}_{F_x}, \mathcal{R}_{F_y}, \mathcal{R}_{F_z} \approx 0$ gives a near-zero phase-locked force ratio. The two estimators use different variance aggregations, but both produce the same ordering.

Sampling uncertainty can be assessed without assuming normally distributed signals by resampling complete demonstrations. If $\Gamma_\chi^{*(b)}$ and $\mathcal{R}_\chi^{*(b)}$ are recomputed from bootstrap resample $b \in \{1, \ldots, N_{\text{boot}}\}$, their percentile intervals are

$$\mathrm{CI}_{0.95}(\Gamma_\chi) = \left[Q_{0.025}\big(\Gamma_\chi^*\big), Q_{0.975}\big(\Gamma_\chi^*\big)\right], \qquad \mathrm{CI}_{0.95}(\mathcal{R}_\chi) = \left[Q_{0.025}\big(\mathcal{R}_\chi^*\big), Q_{0.975}\big(\mathcal{R}_\chi^*\big)\right], \tag{12}$$

where Q_p denotes the empirical p-quantile across bootstrap resamples. This trial-level resampling preserves the temporal dependence within each trajectory. A conservative reliability decision is obtained when the lower confidence limit of every regulated channel exceeds the upper confidence limit of every released channel. Equation (10) reports the observed effect-size margin; confidence intervals should accompany it when the bootstrap results are reported.
The result concluded that: *the demonstrated scalpel path is reproducible; the demonstrated force is not.* The recorded force is dominated by a large, approximately constant offset (consistent with scalpel preload / sensor bias) plus trial-to-trial content that does not repeat; this was checked against the dynamic part of the signal, time alignment and sign convention. The practical implication is that *point-tracking the demonstrated force trajectory is not a well-posed objective.* This is exactly the situation the *minimal-intervention principle* addresses: regulate strongly where the demonstration is consistent (position), and remain compliant where it is not (force). The latter being treated as a constraint and as the interaction outcome of a calibrated material model.

3 Variability-driven variable impedance control

3.1 The impedance control law

The robot is controlled as a virtual mechanical impedance at the scalpel tip. The commanded Cartesian force is

$$\boxed{f \;=\; K(\phi)\,\tilde{x} \;+\; K_I \int_0^t \tilde{x}\,d\sigma \;+\; D(\phi)\,\dot{\tilde{x}}} \tag{13}$$

where,

- $K(\phi)\,\tilde{x}$ is a position-error spring. $K(\phi)\,[\mathrm{N\,m^{-1}}]$ is a diagonal stiffness (one value per Cartesian axis) and $\tilde{x}\,[\mathrm{m}]$ is the position error, so the product has units of force [N]. A larger K pulls the scalpel back to the reference more firmly.
- $K_I \int \tilde{x}\,d\sigma$ is an integral term that removes slow, steady-state position error; $K_I\,[\mathrm{N\,m^{-1}s^{-1}}]$ multiplied by the time-integral of the error $[\mathrm{m \cdot s}]$ again yields [N].
- $D(\phi)\,\dot{\tilde{x}}$ is a velocity-error damper that suppresses oscillation; $D\,[\mathrm{N\,s\,m^{-1}}]$ times $\dot{\tilde{x}}\,[\mathrm{m\,s^{-1}}]$ gives [N].

The damping is tied to the stiffness by a critical-damping rule, $D(\phi) = 2\zeta\sqrt{M\,K(\phi)}$, with damping ratio $\zeta = 0.7$ (a standard, slightly under-damped, fast and non-oscillatory response) and $M\,[\mathrm{kg}]$ an effective mass. The virtual force is realised by joint torques through a damped-least-squares Jacobian transpose, with a posture task in the null space and model-based gravity/Coriolis compensation:

$$\tau \;=\; J^\top\big(JJ^\top + \lambda^2 I\big)^{-1} f \;+\; N\,\tau_{\text{posture}} \;+\; g + C\dot{q}, \qquad N = I - J^\top\big(JJ^\top + \lambda^2 I\big)^{-1} J. \tag{14}$$

where $J\,[\mathrm{m\,rad^{-1}}]$ is the scalpel Jacobian ($\dot{x} = J\dot{q}$); $\left(JJ^\top + \lambda^2 I\right)^{-1}$ is the damped pseudo-inverse that maps the Cartesian force to joint torques while remaining numerically robust near kinematic singularities (λ^2 is a small regulariser); N projects the secondary posture torque τ_{posture} into the null space so it does not disturb the task; and $g + C\dot{q}\,[\mathrm{N\,m}]$ cancel gravity and Coriolis effects so that (13) governs the closed loop.

3.2 Gains learned from demonstration precision

The *novice* work here is that $K(\phi)$ is not chosen manually, nor tabulated or tuned via deterministic regulation but it is the *precision* (inverse variance) of the demonstrations. Intuitively, where the human was very repeatable, that direction matters and should be tracked stiffly; where they were not, the controller should be soft. For each phase ϕ and axis i a precision weight on one shared scale was formed and mapped to stiffness. The calculation is made explicitly as follows. Let N_{demo} be the number of demonstrations and let $y_{\nu,i}(\phi)$ be the value of channel i in demonstration ν at phase ϕ. First, the mean across demonstrations at that same phase is

$$\bar{y}_i(\phi) = \frac{1}{N_{\text{demo}}} \sum_{\nu=1}^{N_{\text{demo}}} y_{\nu,i}(\phi). \tag{15}$$

The cross-trial standard deviation is then

$$s_i(\phi) = \sqrt{\frac{1}{N_{\text{demo}}} \sum_{\nu=1}^{N_{\text{demo}}} \left(y_{\nu,i}(\phi) - \bar{y}_i(\phi)\right)^2}. \tag{16}$$

Thus, $s_i(\phi)$ is small when all demonstrations have nearly the same channel value at phase ϕ, and it is large when their values are widely scattered. The denominator is N_{demo}, rather than $N_{\text{demo}} - 1$, matching the population-standard-deviation convention used in the implementation. Equation (16) is evaluated separately at each of the $N_\phi = 300$ phase samples for X, Y, Z, F_x, F_y, F_z, producing a 300×6 array.

Because the six raw channels have different units and numerical scales, their standard deviations cannot be compared directly. Each channel is therefore normalised by the range of its across-trial mean trajectory,

$$r_i = \max_\phi \bar{y}_i(\phi) - \min_\phi \bar{y}_i(\phi), \qquad \sigma_i(\phi) = \frac{s_i(\phi)}{\max(r_i, \epsilon_r)}, \tag{17}$$

where ϵ_r only protects against a zero range. The resulting $\sigma_i(\phi)$ is a dimensionless cross-trial inconsistency: it measures the scatter at that phase as a fraction of the total demonstrated motion or force range of that channel. Its inverse variance gives the precision

$$p_i(\phi) = \frac{1}{\sigma_i^2(\phi) + \epsilon_\sigma}. \tag{18}$$

Low relative scatter therefore gives high precision, while high relative scatter gives low precision.

To place every axis and channel on one common scale, all $6N_\phi = 1800$ precision values for one material are pooled into the set

$$\mathcal{P} = \{p_i(\phi_\ell) : i \in \{X, Y, Z, F_x, F_y, F_z\},\ \ell = 1, \ldots, N_\phi\}, \qquad p_{\text{ref}} = \text{percentile}_{95}(\mathcal{P}). \tag{19}$$

Thus a separate percentile is not calculated for each axis: the position and force precisions at all 300 phases contribute to the same p_{ref}. Approximately 95% of the pooled precision values are at or below this reference and the upper 5% are above it. The reference is shared across all six channels within a material, but it is recomputed from that material's own demonstrations. For the present data, $p_{\text{ref}} = 42.669$ for cork, 35.895 for PVC, and 18.734 for penoplex.

The final precision weight and its mapping to stiffness are

$$\begin{aligned} w_i(\phi) &= \mathrm{clip}\Big(\frac{p_i(\phi)}{p_{\mathrm{ref}}},\ 0,\ 1\Big) = \mathrm{clip}\Big(\frac{1/(\sigma_i^2(\phi)+\epsilon_\sigma)}{p_{\mathrm{ref}}},\ 0,\ 1\Big), \\ K_i^{\mathrm{pos}}(\phi) &= K_{\mathrm{floor}} + \big(K_{\max} - K_{\mathrm{floor}}\big)\, w_i(\phi). \end{aligned} \tag{20}$$

$\sigma_i^2(\phi)$ is the cross-trial variance of the range-normalised signal at phase ϕ (dimensionless, so axes are comparable); ϵ_σ is a tiny constant avoiding division by zero; p_{ref} is a robust 95^{th}-percentile normaliser *shared across all axes and channels*, so a true imprecise direction receives a true small weight $w_i \to 0$. The weight $w_i \in [0,1]$ then scales the stiffness linearly between a floor K_{floor} [$\mathrm{N\,m^{-1}}$] (kept non-zero so the arm can always execute the motion) and the maximum $K_{\max}$. Precision values at or above p_{ref} are clipped to $w_i = 1$. Using the 95th percentile rather than the absolute maximum prevents a single exceptionally low-variance sample from compressing all other weights towards zero. Figure 1 illustrates the result: with no manual tuning, the cut-depth axis becomes stiff while the lateral and force directions stay compliant — precisely the structure predicted by Section 2.

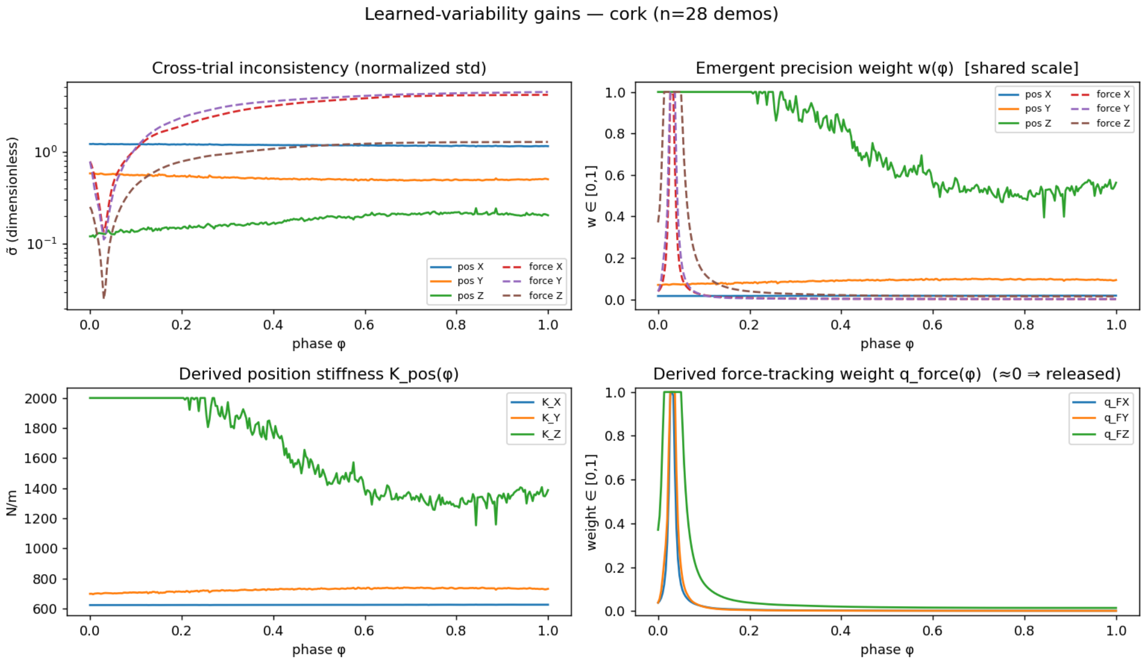

Figure 1: Gains learned from the cork demonstrations. Top-left: cross-trial inconsistency of each channel; top-right: the emergent precision weight $w(\phi)$; bottom: the resulting position stiffness $K^{\mathrm{pos}}(\phi)$ and the (near-zero) force-tracking weight. The depth axis (blue) is regulated stiffly; force is released. *This split emerges from the data, not from tuning.*

The same procedure applied to the other two materials (Fig. 2) yields the *same qualitative structure* — depth regulated, force released — but with *material-specific magnitudes*, because the gains are computed from each material's own demonstration. This is the gain-side counterpart of the per-material force law of Section 6: one principle, three distinct parameter sets.

3.3 The online quadratic program (QP) for the gains

At each control step, the gains are obtained from a small convex quadratic program (QP) that *blends* two objectives and respects two constraints:

$$\boxed{\begin{aligned} K^\star = \ &\arg\min_{K\in\mathbb{R}^3}\ q_f(\phi)\,\|K\odot\tilde{x} - F_d\|_2^2 \ + \ \big(1-q_f(\phi)\big)\,\|K - K^{\mathrm{pos}}(\phi)\|_2^2 \\ &\text{s.t.}\ \ K_{\min} \le K \le K_{\max}, \qquad -F_i^{\mathrm{bound}} \le K_i\,\tilde{x}_i \le F_i^{\mathrm{bound}}\ \ \forall i. \end{aligned}} \tag{21}$$

Term by term:

- $\|K \odot \tilde{x} - F_d\|_2^2$ is the *force-consistency* cost ($\odot$ is element-wise product): it pulls the impedance force $K \odot \tilde{x}$ [N] to match the desired force F_d [N]. Its weight $q_f(\phi) \in [0,1]$ is the *force-precision* weight - small, because force is not reproducible (Sec. 2).

- $\|K - K^{\text{pos}}(\phi)\|_2^2$ pulls the stiffness towards the precision-derived value K^{pos} [$\mathrm{N\,m^{-1}}$] of (20), weighted by $1 - q_f$. Because $q_f \to 0$, this term dominates and the QP returns $K \approx K^{\text{pos}}$: the controller tracks the precision-weighted position skill.

- The box constraint $K_{\min} \leq K \leq K_{\max}$ keeps the stiffness in a safe physical range.

- The *force-saturation* constraint $|K_i \tilde{x}_i| \leq F_i^{\text{bound}}$ guarantees the commanded force on each axis never exceeds the demonstrated force limit F_i^{bound} [N] — i.e. the desired forces act as *constraints*, as required.

The damping follows from $K^\star$ by the critical-damping rule. The QP is tiny (three variables) and solves in well under a millisecond (OSQP), so it runs in the real-time loop.

highlighting the contribution of the learned variance structure. To establish that the performance gain originates from the *specific* learned precision structure, rather than from gain adaptation per se, evaluation of the identical task and controller under four gain assignments is presented: (i) the identified precision; (ii) *inverted* precision (regulating directions that were inconsistent and relaxing those that were consistent); (iii) precision *permuted* across axes; and (iv) a baseline that tracks the demonstrated force directly. The metric is the *cut-depth error* — the task-critical, demonstration-regulated axis — evaluated over 12 seeds with randomised contact parameters (Fig. 3, Table 3).

Let $w(\phi) = [w_1(\phi), w_2(\phi), w_3(\phi)]^\top \in [0,1]^3$ be the identified position-precision vector of (20), and define the affine normalisation of the Cartesian stiffness vector by

$$u(\phi) = \frac{K^{\text{pos}}(\phi) - K_{\min}\mathbf{1}}{K_{\max} - K_{\min}} \in [0,1]^3, \qquad K^{\text{pos}}(\phi) = K_{\min}\mathbf{1} + (K_{\max} - K_{\min})u(\phi). \tag{22}$$

Here $K^{\text{pos}} = [K_1^{\text{pos}}, K_2^{\text{pos}}, K_3^{\text{pos}}]^\top$ is the Cartesian stiffness vector, $\mathbf{1} \in \mathbb{R}^3$ is the all-ones vector, and the scalar denominator is the full admissible stiffness range. Thus $u_i = 0$ and $u_i = 1$ correspond respectively to $K_i^{\text{pos}} = K_{\min}$ and $K_i^{\text{pos}} = K_{\max}$; u is a dimensionless coordinate representation of the commanded stiffness rather than an additional controller state. Substitution of the learned gain law (20) into (22) gives

$$\begin{aligned} u_w(\phi) &= \eta_0\mathbf{1} + \eta_1 w(\phi), \\ \eta_0 &= \frac{K_{\text{floor}} - K_{\min}}{K_{\max} - K_{\min}}, \\ \eta_1 &= \frac{K_{\max} - K_{\text{floor}}}{K_{\max} - K_{\min}}. \end{aligned} \tag{23}$$

Accordingly, u_w denotes the normalized Cartesian stiffness induced by the precision weights obtained from the demonstrations.

The controlled comparison uses the exact transformations

$$\begin{aligned} u^{\text{true}}(\phi) &= u_w(\phi), \\ u^{\text{inv}}(\phi) &= \mathbf{1} - u_w(\phi), \qquad \Pi(\phi)^\top \Pi(\phi) = I, \\ u^{\text{perm}}(\phi) &= \Pi(\phi) u_w(\phi), \end{aligned} \tag{24}$$

The matched condition applies each learned component to its identified Cartesian axis. The inverted condition acts componentwise and satisfies

$$u_i^{\text{inv}}(\phi) = 1 - u_{w,i}(\phi), \qquad u_i^{\text{inv}} - u_j^{\text{inv}} = -\big(u_{w,i} - u_{w,j}\big), \tag{25}$$

thereby reversing the component ordering about the midpoint of the admissible normalized stiffness interval. The permuted condition preserves the instantaneous multiset of components of u_w but changes their Cartesian-axis assignment. These transformations distinguish sensitivity to the learned ordering from sensitivity to the learned axis association. The direct-force condition is not an allocation transformation in this space: it disables the learned-gain mode and applies the fixed force-QP weighting of the baseline controller.

Write the identified allocation as

$$u_w(\phi) = \begin{bmatrix} u_{w,x}(\phi) & u_{w,y}(\phi) & u_{w,z}(\phi) \end{bmatrix}^\top . \tag{26}$$

For a bijection $\pi_\phi : \{1,2,3\} \to \{1,2,3\}$, define

$$[\Pi(\phi)]_{jk} = \begin{cases} 1, & k = \pi_\phi(j), \\ 0, & \text{otherwise}, \end{cases} \qquad j,k \in \{1,2,3\}. \tag{27}$$

It follows that

$$[\Pi(\phi)u_w(\phi)]_j = u_{w,\pi_\phi(j)}(\phi), \qquad \Pi(\phi)^\top \Pi(\phi) = I, \qquad \Pi(\phi)^{-1} = \Pi(\phi)^\top . \tag{28}$$

Consequently, $\Pi(\phi)$ preserves the Euclidean norm and the multiset of components of $u_w(\phi)$ while changing their Cartesian-axis assignment. The dependence on ϕ permits the assignment to vary over task phase. In the implemented shuffled condition, ϕ determines the pseudorandom seed used to generate π_ϕ. This comparison isolates the effect of the learned axis assignment while preserving the instantaneous set of stiffness allocations.

Proposition 2 (Uniqueness of the matched allocation under the precision-deviation functional)**.** *Consider the precision-allocation functional*

$$\mathcal{J}[u] = \frac{1}{2}\int_0^1 \big(u(\phi) - u_w(\phi)\big)^\top H(\phi)\big(u(\phi) - u_w(\phi)\big)\, d\phi, \qquad H(\phi) = H(\phi)^\top \succ 0. \tag{29}$$

Among all measurable allocations $u : [0,1] \to [0,1]^3$, the matched assignment $u^{\text{true}} = u_w$ is the unique minimiser of (29), up to sets of zero measure. The excess costs of the inverted and permuted structures are

$$\mathcal{G}_{\text{inv}} = \frac{1}{2}\int_0^1 (\mathbf{1} - 2u_w)^\top H(\mathbf{1} - 2u_w)\, d\phi, \tag{30}$$

$$\mathcal{G}_{\text{perm}} = \frac{1}{2}\int_0^1 (\Pi u_w - u_w)^\top H(\Pi u_w - u_w)\, d\phi. \tag{31}$$

Both are strictly positive whenever the transformation changes a task-weighted component of u_w on a phase interval of non-zero measure.

Proof. Since $H(\phi) \succ 0$, its smallest eigenvalue $h_{\min}(\phi) = \lambda_{\min}(H(\phi))$ is positive and

$$\mathcal{J}[u] \geq \frac{1}{2}\int_0^1 h_{\min}(\phi)\, \|u(\phi) - u_w(\phi)\|_2^2 \, d\phi \geq 0. \tag{32}$$

The lower bound is attained by $u = u_w$, for which $\mathcal{J}[u_w] = 0$. Equality for another allocation requires $u(\phi) = u_w(\phi)$ almost everywhere, which proves uniqueness. Substitution of $u^{\text{inv}} = \mathbf{1} - u_w$ and $u^{\text{perm}} = \Pi u_w$ into (29) gives (30) and (31). Positive definiteness of H makes either integral strictly positive when its difference vector is non-zero on a set of non-zero measure. □

Define the allocation error

$$e_u(\phi) = u(\phi) - u_w(\phi) = \begin{bmatrix} e_x(\phi) & e_y(\phi) & e_z(\phi) \end{bmatrix}^\top . \tag{33}$$

Then the integrand of (29) is the quadratic form

$$\frac{1}{2}e_u^\top H e_u = \frac{1}{2}\sum_{j=1}^{3}\sum_{k=1}^{3} H_{jk}(\phi)e_j(\phi)e_k(\phi). \tag{34}$$

Here $H : [0,1] \to \mathbb{S}^3_{++}$ is a measurable, phase-dependent task-relevance metric on the allocation space. Its diagonal entries weight axis-specific allocation errors, whereas its off-diagonal entries encode cross-axis coupling. The condition $H(\phi) \succ 0$ means

$$v^\top H(\phi)v > 0 \quad \text{for every non-zero } v \in \mathbb{R}^3, \tag{35}$$

and is the condition required for the uniqueness result in Proposition 2. For the cutting task, greater depth-axis relevance may be represented through the corresponding entries of $H(\phi)$.

The matrix $H(\phi)$ is an assumed metric defining the quadratic allocation functional; it is not estimated from the demonstrations, numerically specified in the experiments, or evaluated by the controller. Consequently, Proposition 2 establishes only that u_w uniquely minimizes the defined weighted deviation from u_w. It does not establish optimality with respect to cutting error, interaction force, or a general closed-loop performance functional. The physical consequence of the matched allocation is instead evaluated by the paired closed-loop experiment below.

For controller condition c and randomised-physics seed s, the reported cut-depth metric is

$$E^z_{s,c} = 10^3\sqrt{\frac{1}{t_{f,s}}\int_0^{t_{f,s}} \left(x^{(s,c)}_{d,z}(t) - x^{(s,c)}_z(t)\right)^2 dt} \quad \text{[mm]}, \tag{36}$$

where $t_{f,s}$ is the episode duration, $x_{d,z}$ is desired scalpel depth, x_z is measured depth, and the factor 10^3 converts metres to millimetres. The same randomised mass, friction, contact parameters and initial perturbation are used for all conditions within seed s, so the paired difference

$$G_{s,c} = E^z_{s,c} - E^z_{s,\text{true}} \tag{37}$$

removes seed-specific physics shared by the two runs. Positive $G_{s,c}$ means that condition c has larger cut-depth error than the identified precision assignment.

Gain assignment	depth RMSE (mm)	identified reduction	Wilcoxon p
Identified precision	$\mathbf{10.9 \pm 2.5}$	—	—
Permuted precision	14.7 ± 2.9	+26%	0.001
Inverted precision	20.7 ± 4.0	+47%	0.001
Direct force-tracking	20.9 ± 5.9	+48%	0.001

Table 3: The identified precision attains the lowest cut-depth error on all 11 stable seeds and reduces error by 26–48% relative to the three comparator assignments ($p = 0.001$, the smallest value attainable by the paired Wilcoxon test at this sample size). This isolates the identified *variance structure*, rather than gain adaptation in general, as the determinant of task performance.

For the $n_{\text{pair}} = 11$ seeds that were stable in every paired condition, the table reports

$$\bar{E}_c = \frac{1}{n_{\text{pair}}}\sum_{s=1}^{n_{\text{pair}}} E^z_{s,c}, \qquad s_c = \sqrt{\frac{1}{n_{\text{pair}}-1}\sum_{s=1}^{n_{\text{pair}}}\left(E^z_{s,c} - \bar{E}_c\right)^2}, \tag{38}$$

where $\bar{E}_c$ and s_c are respectively the sample mean and sample standard deviation for condition c. The percentage shown in the third column is the error reduction achieved by the identified assignment relative to comparator c,

$$R_c = 100\,\frac{\bar{E}_c - \bar{E}_{\text{true}}}{\bar{E}_c} \quad [\%], \tag{39}$$

which gives 26%, 47% and 48% for the permuted, inverted and direct-force conditions. This denominator explains why these percentages are reductions relative to each comparator rather than percentage increases relative to the identified mean.

Proposition 3 (Paired dominance in the evaluated seeds)**.** *For each of the three comparator conditions, the evaluated stable pairs satisfy $G_{s,c} > 0$ for all $s \in \{1, \dots, 11\}$. The exact two-sided paired Wilcoxon signed-rank test therefore gives*

$$p_{\text{exact}} = 2^{1-n_{\text{pair}}} = 2^{-10} = 0.0009765625 \approx 0.001. \tag{40}$$

Proof. Let ϱ_s be the rank of $|G_{s,c}|$ among the n_{pair} non-zero paired differences and define the negative signed-rank statistic

$$W_c^- = \sum_{s=1}^{n_{\text{pair}}} \varrho_s \nVdash_{\{G_{s,c}<0\}}. \tag{41}$$

Because every observed difference is positive, $W_c^- = 0$. Under the Wilcoxon null hypothesis of a distribution symmetric about zero, each of the $2^{n_{\text{pair}}}$ sign assignments is equally likely. Only the all-positive and all-negative assignments are at least as extreme as the observed one, so the two-sided exact probability is $2/2^{n_{\text{pair}}} = 2^{1-n_{\text{pair}}}$. Substitution of $n_{\text{pair}} = 11$ yields (40). □

Proposition 3 is an exact statement about the evaluated paired seeds and rejects the zero-median-difference null at the reported sample size. Together with Proposition 2, it links the analytical consequence of matching the learned precision allocation to the observed closed-loop reduction in the task-critical error, without treating the finite simulation sample as a universal proof for all contact conditions.

4 State-driven task parametrization

The reference is a function of a *phase* variable $\phi \in [0, 1]$ to be decoupled from direct time dependency. This separates *what* the skill is from *how fast* it runs, which allows the same prior be executed at different speeds. The *novice* contribution here is *how* ϕ advances: not by a but by a correction driven by the robot's *measured state* instead of a fixed time cycle that is affected by many factors like processor power, hardware, memory and process scheduling, system specifications or other real time application issues which might break the generalizability across different conditions. At every step, the phase $\hat{\phi}$ that best matches the current state to the learned trajectory over a short forward window, is evaluated then correct ϕ towards it:

$$\hat{\phi} = \arg \min_{\varphi \in [\phi,\, \phi + \ell_\phi]} \left[\frac{\|x_{\text{ref}}(\varphi) - x\|}{\rho_x} + \nVdash_{\text{contact}} \frac{\big|\, \|F_{\text{ref}}(\varphi)\| - \|F\| \,\big|}{\rho_F} \right], \qquad \phi \leftarrow \text{clip}\big(\phi + \dot{\phi}_{\text{nom}} + k_\phi(\hat{\phi} - \phi)\big). \tag{42}$$

The bracketed cost compares where the robot *is* to where each candidate phase φ assumes it *should be*; ℓ_ϕ is the forward phase-search horizon. The first term is the position mismatch $\|x_{\text{ref}}(\varphi) - x\|$ [m] normalised by a scale ρ_x [m]; the second, active only in contact ($\nVdash_{\text{contact}}$), compares force magnitudes normalised by ρ_F [N] — force is the more informative cue once the scalpel is engaged. The update mixes the nominal clock rate $\dot{\phi}_{\text{nom}}$ [s^{-1}] with a proportional correction of gain $k_\phi \in [0, 1]$ towards the state estimate $\hat{\phi}$. Setting $k_\phi = 0$ recovers a pure time clock; with $k_\phi > 0$ the phase *tracks the robot's actual progress* — it slows or pauses if the robot is held back and speeds up if it runs ahead. Searching only forward ($\varphi \geq \phi$) guarantees monotone progress, so the skill never regresses to an earlier stage. This provides speed-adaptability and robustness to disturbance execution.

5 Stability and passivity analysis

The scalpel obeys the task-space dynamics $\Lambda\ddot{x} + C\dot{x} + g = f + f_{\text{ext}}$, where Λ [kg] is the Cartesian inertia and f_{ext} [N] the environment force. Substituting the control law (13) (with gravity/Coriolis cancelled by (14)) gives the *error dynamics*

$$\Lambda\ddot{\tilde{x}} + \big(C + D(\phi)\big)\dot{\tilde{x}} + K(\phi)\,\tilde{x} = f_{\text{ext}}, \tag{43}$$

a (time-varying) mass–spring–damper in the tracking error driven by the contact force.

5.1 Passivity — the controller cannot create energy

Proposition 4 (Continuous-time tank passivity)**.** *At the Cartesian contact port, the input–output map $f_{ext} \mapsto \dot{x}$ is passive under the following tank dynamics: With the storage function (an energy, in joules) $\mathcal{S} = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x} + T$ and a non-negative* energy tank $T \geq T_{\min}$ *whose dynamics collect the power dissipated by damping and compensate the power injected by the time-varying gains and moving reference,*

$$\dot{T} = \underbrace{\dot{\tilde{x}}^\top D\,\dot{\tilde{x}}}_{\textit{collected}\ [\mathrm{W}]} - \underbrace{P_{inj}}_{[\mathrm{W}]}, \qquad P_{inj} = \tfrac{1}{2}\tilde{x}^\top \dot{K}\,\tilde{x} + \tilde{x}^\top K\,\dot{x}_d, \tag{44}$$

the loop satisfies $\dot{\mathcal{S}} \leq f_{ext}^\top \dot{x}$ as long as $T > T_{\min}$.

Interpretation: $\mathcal{S}$ is the total stored energy (kinetic + elastic + tank). The two indefinite, potentially energy-injecting terms — the rate of change of stored elastic energy from a rising stiffness ($\frac{1}{2}\tilde{x}^\top \dot{K}\tilde{x}$) and the power supplied to a moving reference ($\tilde{x}^\top K\dot{x}_d$), both in watts — are drawn from the tank, which is recharged by the power dissipated through damping. When the tank reaches its lower bound, stiffness growth is inhibited ($\dot{K} \leq 0$) and the phase is frozen ($\dot{\phi} \to 0$), so the controller cannot generate energy, instead, it reduces the motion velocity to avoid instant change of dynamics. This is the continuous-time passivity argument used to design the implementation.

5.2 Implemented torque-port passivity

The simulation controller enforces the tank constraint at the joint torque port, where the mechanical power supplied by the actuators is $P_\tau = \dot{q}^\top \tau$. At every control step the nominal torque

$$\tau_{\text{nom}} = \tau_{\text{task}} + \tau_{\text{null}} + \tau_{\text{bias}}, \qquad \tau_{\text{task}} = J^\top (JJ^\top + \lambda^2 I)^{-1} f, \qquad \tau_{\text{null}} = N\tau_{\text{posture}}, \qquad \tau_{\text{bias}} = g + C\dot{q}$$

is assembled from the task, null-space and model-compensation components. The task torque τ_{task} [N m] maps the Cartesian impedance command f [N] to the joints through the damped Jacobian inverse, where $J \in \mathbb{R}^{3\times n}$ satisfies $\dot{x} = J\dot{q}$, λ^2 is the damped-least-squares regulariser, I is the identity matrix and n is the number of joints. The posture command τ_{posture} [N m] attracts the arm towards its home configuration and damps joint motion. Its projection $\tau_{\text{null}} = N\tau_{\text{posture}}$, with $N = I - J^\top (JJ^\top + \lambda^2 I)^{-1} J$, acts in the redundant null space and therefore satisfies $JN \approx 0$ without materially disturbing the Cartesian cutting task. The bias term τ_{bias} [N m] contains the gravity torque g and the Coriolis/centrifugal torque $C\dot{q}$. Their sum τ_{nom} [N m] is the nominal joint-torque command prior to passivity enforcement.

The nominal command is projected onto the half-space of admissible torques:

$$\boxed{\tau_{\text{safe}} = \arg\min_{\tau} \|\tau - \tau_{\text{nom}}\|_2^2 \quad \text{s.t.} \quad \dot{q}^\top \tau \leq P_{\max}, \qquad P_{\max} = \frac{T - T_{\min}}{h_c}.} \tag{45}$$

Here τ_{safe} [N m] is the projected actuator command, q [rad] and $\dot{q}$ [rad s^{-1}] are the joint position and velocity, and $\dot{q}^\top \tau$ [W] is the mechanical power supplied at the joint port. The scalar $P_{\max}$ [W] is the power available from the usable tank energy $T - T_{\min}$ over one control period h_c [s]. Consequently, the constraint bounds positive energy delivery by the controller, while negative joint power corresponds to energy absorption. The quadratic objective retains the nominal control action as closely as possible in the Euclidean torque norm. If $\dot{q}^\top \tau_{\text{nom}} \le P_{\max}$, then $\tau_{\text{safe}} = \tau_{\text{nom}}$. Otherwise the Euclidean projection has the closed form

$$\tau_{\text{safe}} = \tau_{\text{nom}} - \frac{\dot{q}^\top \tau_{\text{nom}} - P_{\max}}{\dot{q}^\top \dot{q} + \epsilon_\tau}\, \dot{q}, \tag{46}$$

where $\epsilon_\tau > 0$ is a small numerical regulariser. The correction lies along $\dot{q}$, the direction that changes instantaneous joint power, and is therefore the minimum-norm torque modification that satisfies the power bound. This expression is also used as a fallback if the numerical QP does not return a solution.

Before the torque projection, the tank state and the phase gate are updated as

$$T_{k+1} = \operatorname{clip}\big(T_k + \big(P_{\text{dis}} - P_{\text{inj}}\big)h_c,\, T_{\min},\, T_{\max}\big)\,, \qquad \gamma_T = \tanh\!\left(\frac{T_{k+1} - T_{\min}}{T_{\text{ref}}}\right), \tag{47}$$

where T_k [J] and T_{k+1} [J] are the tank energies at consecutive samples, $T_{\min}$ [J] is the protected lower bound, $T_{\max}$ [J] is the storage ceiling, and $\operatorname{clip}(\cdot)$ enforces these bounds. The collected damping power is $P_{\text{dis}} = \dot{\tilde{x}}^\top D \dot{\tilde{x}}$ [W]. The withdrawn power $P_{\text{inj}} = P_{\text{gain}} + P_{\text{ref}}$ comprises the positive power P_{gain} associated with increasing $K(\phi)$ and the positive power P_{ref} associated with motion of $x_d(\phi)$. The reference energy T_{ref} [J] sets the transition scale of the phase gate: $\gamma_T \in [0, 1]$ approaches zero as the usable tank energy $T_{k+1} - T_{\min}$ vanishes and modulates the phase rate according to $\dot{\phi} = \gamma_T \dot{\phi}_{\text{nom}}$. Increasing T_{ref} makes this transition more gradual over the available energy range.

The resulting closed-loop sequence couples the timing, impedance and joint-torque layers. The available tank energy first determines γ_T and hence the phase rate. The updated phase selects x_d, $\dot{x}_d$, $K(\phi)$ and $D(\phi)$, from which the impedance law produces f. The task mapping, null-space posture term and dynamic compensation then form τ_{nom}. After collection of P_{dis} and withdrawal of P_{inj}, the updated tank state determines $P_{\max}$, and the projection produces τ_{safe} satisfying $\dot{q}^\top \tau_{\text{safe}} \le P_{\max}$. Phase gating therefore reduces the energy demand generated by reference evolution, while the torque projection enforces the joint-port power bound at every sample.

Controller	completion	min. tank (J)	budget violation	max margin (W)
No tank	100%	0.001	12.0%	+5876
Tank projection	100%	0.907	0.0%	0
Tank + phase gate	100%	0.921	0.0%	0

Table 4: Torque-port passivity validation under a positive-power injection. The tanked controllers complete the cut and never exceed the available energy budget.

This is the empirical passivity proof artifact used in this sudy. The reconstructed Cartesian residual $\dot{V} - f_{\text{ext}}^\top \dot{x}$ is still plotted as a diagnostic in Fig. 5, but it depends on finite differences, force-estimator sign conventions and an approximate Cartesian inertia; the implemented guarantee is therefore stated and validated at the torque port $\dot{q}^\top \tau$.

5.3 Exponential stability — the error decays at a guaranteed rate

The stability argument is first established for nominal regulation and is then extended to phase-varying gains and bounded perturbations.

Assumption 1 (Nominal regulation conditions)**.** *The compensated Cartesian inertia satisfies $\Lambda = \Lambda^\top \succ 0$ and is constant over the local analysis interval; the frozen gains satisfy $K = K^\top \succ 0$ and $D = D^\top \succ 0$; gravity and Coriolis terms are compensated; and the nominal result is evaluated with zero external input and zero unmodelled reference acceleration.*

Under Assumption 1, defining $e = \tilde{x}$, $v = \dot{\tilde{x}}$ and $z = [e^\top\ v^\top]^\top$, the error system becomes

$$\dot{z} = Az, \qquad A = \begin{bmatrix} 0 & I \\ -\Lambda^{-1}K & -\Lambda^{-1}D \end{bmatrix}. \tag{48}$$

The matrix $\Lambda\,[\mathrm{kg}]$ is the Cartesian inertia, $K\,[\mathrm{N\,m^{-1}}]$ and $D\,[\mathrm{N\,s\,m^{-1}}]$ are the frozen stiffness and damping matrices, and A is the corresponding state matrix. The mechanical energy $V_0 = \frac{1}{2}v^\top \Lambda v + \frac{1}{2}e^\top K e$ satisfies only $\dot{V}_0 = -v^\top D v \le 0$. A strict decay rate in both e and v is obtained by adding a position–velocity cross-term:

$$V(z) = \tfrac{1}{2}v^\top \Lambda v + \tfrac{1}{2}e^\top K e + \beta e^\top \Lambda v = \tfrac{1}{2}z^\top P z, \qquad P = \begin{bmatrix} K & \beta\Lambda \\ \beta\Lambda & \Lambda \end{bmatrix}, \qquad \beta > 0. \tag{49}$$

Here $\beta\,[\mathrm{s^{-1}}]$ weights the cross-term. By the Schur complement,

$$P \succ 0 \iff K - \beta^2\Lambda \succ 0, \tag{50}$$

for which the scalar condition $\beta^2 < \lambda_{\min}(K)/\lambda_{\max}(\Lambda)$ is sufficient. Hence, with $\underline{p} = \lambda_{\min}(P)$ and $\overline{p} = \lambda_{\max}(P)$,

$$\frac{\underline{p}}{2}\,\|z\|^2 \le V(z) \le \frac{\overline{p}}{2}\,\|z\|^2. \tag{51}$$

Theorem 1 (Exponential stability)**.** *For the nominal system* (48)*, suppose $P \succ 0$ and*

$$Q = \begin{bmatrix} 2\beta K & \beta D \\ \beta D & 2(D - \beta\Lambda) \end{bmatrix} \succ 0. \tag{52}$$

Then the origin $z = 0$ is globally exponentially stable. In particular, with $\underline{q} = \lambda_{\min}(Q)$,

$$\dot{V} \le -\alpha V, \qquad \alpha = \frac{\underline{q}}{\overline{p}}, \qquad \|z(t)\| \le \sqrt{\frac{\overline{p}}{\underline{p}}}\, e^{-\alpha t/2}\,\|z(0)\|. \tag{53}$$

Proof. Using $\Lambda\dot{v} = -Dv - Ke$, differentiation of (49) gives

$$\begin{aligned} \dot{V} &= v^\top \Lambda \dot{v} + e^\top K v + \beta v^\top \Lambda v + \beta e^\top \Lambda \dot{v} \\ &= -v^\top (D - \beta\Lambda) v - \beta e^\top K e - \beta e^\top D v = -\tfrac{1}{2} z^\top Q z. \end{aligned} \tag{54}$$

The cancellation of $-v^\top K e + e^\top K v$ removes the conservative spring-power exchange. From $Q \succ 0$ and (51),

$$\dot{V} \le -\frac{\underline{q}}{2}\,\|z\|^2 \le -\frac{\underline{q}}{\overline{p}} V = -\alpha V.$$

Integration gives $V(t) \le e^{-\alpha t} V(0)$, and substitution of the upper and lower quadratic bounds in (51) yields (53). □

Condition (52) can be checked directly from the learned gains and Cartesian inertia at each phase. Its Schur-complement form,

$$D - \beta\Lambda - \frac{\beta}{4} D K^{-1} D \succ 0, \tag{55}$$

also shows why $D - \beta\Lambda \succ 0$ alone is insufficient: the mixed term $-\beta e^\top D v$ must also be dominated. The computable rate $\alpha = \lambda_{\min}(Q)/\lambda_{\max}(P)$ expresses the joint influence of inertia, stiffness, damping and the selected cross-term coefficient.

Time-varying gains and moving-reference perturbations. For $K = K(\phi)$, $D = D(\phi)$ and an aggregated Cartesian perturbation δ containing external contact input, imperfect model compensation and uncompensated reference acceleration, the derivative becomes

$$\dot{V} = -\tfrac{1}{2} z^\top Q(\phi) z + \tfrac{1}{2} e^\top \dot{K}(\phi) e + (v + \beta e)^\top \delta, \qquad \dot{K}(\phi) = K'(\phi)\dot{\phi}. \tag{56}$$

Corollary 1 (Uniform exponential stability with variable gains)**.** *Suppose $P(\phi) \succ 0$ and $Q(\phi) \succ 0$ uniformly for $\phi \in [0, 1]$, and define*

$$q_* = \inf_{\phi\in[0,1]} \lambda_{\min}\big(Q(\phi)\big), \qquad \kappa = \sup_{\phi\in[0,1]} \left\| \dot{K}(\phi) \right\|_2, \qquad \mu = q_* - \kappa. \tag{57}$$

If $\mu > 0$, the unforced time-varying system is uniformly exponentially stable. Since $\left\| \dot{K}(\phi) \right\|_2 \le \|K'(\phi)\|_2 \, |\dot{\phi}|$, a sufficient phase-rate condition is

$$|\dot{\phi}| < \frac{q_*}{\sup\limits_{\phi\in[0,1]} \|K'(\phi)\|_2}. \tag{58}$$

Proof. With $\delta = 0$, equations (56) and (57) give

$$\dot{V} \le -\tfrac{1}{2}\big(q_* - \kappa\big) \|z\|^2 = -\tfrac{1}{2}\mu \|z\|^2 .$$

Uniform positive lower and upper bounds on $P(\phi)$ imply exponential decay by the comparison argument used in Theorem 1. Condition (58) ensures $\kappa < q_*$ and therefore $\mu > 0$. □

The tank gate (47) reduces $\dot{\phi}$ as usable energy decreases and therefore reduces $\dot{K}$ and the moving-reference power demand. The torque projection (45) independently enforces the joint-port power constraint. Passivity follows from the energy accounting, whereas uniform exponential stability additionally requires the matrix margin (57); the two properties are related through phase gating but are not identical claims.

Proposition 5 (Practical exponential stability under bounded perturbations)**.** *Let the conditions of Corollary 1 hold, define $g_\beta = \sqrt{1 + \beta^2}$, and suppose that $\sup_{s\ge 0} \|\delta(s)\| < \infty$. Then*

$$\dot{V} \le -\frac{\mu}{4} \|z\|^2 + \frac{g_\beta^2}{\mu} \|\delta\|^2 \le -aV + \frac{g_\beta^2}{\mu} \|\delta\|^2 , \qquad a = \frac{\mu}{2\overline{p}}, \tag{59}$$

where $\overline{p} = \sup_\phi \lambda_{\max}(P(\phi))$. Integration gives

$$V(t) \le e^{-at} V(0) + \frac{g_\beta^2}{a\mu} \left(1 - e^{-at}\right) \sup_{0\le s\le t} \|\delta(s)\|^2 . \tag{60}$$

Accordingly, the state converges exponentially to a residual set whose radius is proportional to $\sup \|\delta\|$; when $\delta = 0$, this set collapses to the origin.

Proof. The perturbation term in (56) satisfies $|(v + \beta e)^\top \delta| \le g_\beta \|z\| \, \|\delta\|$. Young's inequality with the uniform margin μ yields (59); integration of the scalar differential inequality gives (60). □

Experimental interpretation. Theorem 1 provides the frozen-gain nominal result, Corollary 1 states the uniform margin required while the learned gains evolve with phase, and Proposition 5 accounts for bounded workpiece motion, contact forces and compensation error. Figure 5 reports the corresponding empirical diagnostics. The cut-depth error remains small and the storage remains bounded. The fitted decay of the full Cartesian error is interpreted cautiously because low-precision axes are intentionally assigned low stiffness and therefore have smaller contraction margins.

Implementation status. The live tank state determines the torque-level budget in (45), while phase progression is gated by (47). The positive-power perturbation validation in Fig. 4 and Table 4 evaluates the implemented passivity constraint. Equations (52)–(60) separately state the nominal, uniform time-varying and perturbed stability conditions against which the logged trajectories are assessed.

6 Material interaction model

In simulation the controller tracked position well, yet the contact force at first failed to match the demonstrated force. The stock simulated *material model* presents both the scalpel blade and the workpiece as rigid box geometries; hence, overlapping boxes in a rigid-contact simulator produce solver-dependent *collision forces*, unlike the physical nature of the real cutting force. Their magnitude is set by contact penetration, time step and solver parameters, and their direction by the instantaneous collision normal. A controller cannot make such a contact respect a demonstrated cutting law, because the measured "force" is a geometric contact artefact rather than a material-specific *fracture force*. Figure 6 shows this failure mode in a matched comparison: the same controller, trajectory, gains and cork workpiece are used, and only the contact model is toggled.

Proposition 6 (Why box-on-box contact is not a cutting model)**.** *Let the desired cutting interaction be a material law* $F_d(\phi) = \bar{F}(d(\phi))\, r(\phi)$ *with bounded magnitude* $0 \le \bar{F}(d) \le F^{\mathrm{cut}}_{\mathrm{mat}}$ *and unit reaction direction* $r(\phi)$ *learned from the demonstration. A rigid box-on-box MuJoCo contact generates instead*

$$F_{\mathrm{box}} = F_{\mathrm{solver}}(q, \dot{q}, h_{\mathrm{sim}}, \texttt{solref}, \texttt{solimp}, \mathcal{C}(q), \mu_c), \tag{61}$$

where h_{sim} *is the simulator integration step,* $\mathcal{C}(q)$ *is the active contact set and* μ_c *denotes the friction parameters. In general there exists no depth-only map* $\mathcal{F} : \mathbb{R}_{\ge 0} \to \mathbb{R}^3$ *that is invariant to these numerical and geometric quantities and satisfies simultaneously* $F_{\mathrm{box}} = \mathcal{F}(d) = \bar{F}(d) r(\phi)$ *along the demonstrated cut.*

Proof. Assume such an invariant $\mathcal{F}(d)$ exists. Consider two simulator configurations with the same blade pose and hence the same penetration d, but with distinct solver settings $\vartheta_1 = (h_{\mathrm{sim},1}, \texttt{solref}_1, \texttt{solimp}_1)$ and $\vartheta_2 = (h_{\mathrm{sim},2}, \texttt{solref}_2, \texttt{solimp}_2)$. The normal reaction in a penalty or constraint-regularised contact has the generic dependence

$$F_{n,i} = \Psi(d, \dot{d}; \vartheta_i), \qquad i \in \{1, 2\}, \tag{62}$$

and changing the effective contact stiffness, damping or discrete step gives $\Psi(d, \dot{d}; \vartheta_1) \ne \Psi(d, \dot{d}; \vartheta_2)$ for some $(d, \dot{d})$. Thus identical d produces different force magnitudes, which contradicts the assumed depth-only invariance.

Independently, the rigid-contact force satisfies the friction-cone restriction

$$F_{\mathrm{box}} = F_n n_c + F_t, \qquad F_n \ge 0, \quad n_c^\top F_t = 0, \quad \|F_t\| \le \mu_c F_n, \tag{63}$$

where n_c is a normal selected by the active box feature. At a face or edge transition, $\mathcal{C}(q)$ and n_c can change while d remains continuous. The resulting force direction is therefore determined by collision geometry and need not equal the demonstrated $r(\phi)$. Since equality with the cutting law requires both

$$\|F_{\mathrm{box}}\| = \bar{F}(d) \quad \text{and} \quad \frac{F_{\mathrm{box}}}{\|F_{\mathrm{box}}\|} = r(\phi),$$

failure of either condition disproves $F_{\mathrm{box}} = \bar{F}(d) r(\phi)$. Hence the rigid collision map cannot supply the missing material-specific constitutive relation. □

The rigid contact is replaced with a *per-material cutting-force law.* Physically, cutting force rises elastically with penetration and then *plateaus* at the material's cutting strength (the blade fractures/cuts the material). Let $p_b(t) \in \mathbb{R}^3$ be the blade-tip position, $p_s(t) \in \mathbb{R}^3$ a point on the live workpiece surface, and $n_s(t) \in \mathbb{R}^3$, $\|n_s\| = 1$, its outward normal. The instantaneous penetration is

$$d(t) = \Big[-n_s(t)^\top \big(p_b(t) - p_s(t)\big)\Big]_+, \qquad [\xi]_+ = \max(\xi, 0). \tag{64}$$

Using the current $p_s(t)$ rather than its initial value preserves the penetration definition for a moving workpiece. Let $\hat{F}_d(\phi) = F_d(\phi)/(\|F_d(\phi)\| + \varepsilon_F)$ denote the regularised demonstrated force direction. The reaction applied to the blade is

$$\boxed{\begin{aligned} f_{\mathrm{mat}}(d;\theta_{\mathrm{mat}}) &= \min\big(k_{\mathrm{mat}} d, F^{\mathrm{cut}}_{\mathrm{mat}}\big), \\ F_{\mathrm{react}}(d,\phi) &= -f_{\mathrm{mat}}(d;\theta_{\mathrm{mat}})\,\hat{F}_d(\phi), \\ \theta_{\mathrm{mat}} &= \big(k_{\mathrm{mat}}, F^{\mathrm{cut}}_{\mathrm{mat}}\big), \qquad d_c = \frac{F^{\mathrm{cut}}_{\mathrm{mat}}}{k_{\mathrm{mat}}}. \end{aligned}} \tag{65}$$

Here $k_{\mathrm{mat}}\,[\mathrm{N\,m^{-1}}]$ is the pre-fracture loading stiffness, $F^{\mathrm{cut}}_{\mathrm{mat}}\,[\mathrm{N}]$ is the cutting-force plateau, and $d_c\,[\mathrm{m}]$ is the transition depth. Equivalently,

$$f_{\mathrm{mat}}(d) = \begin{cases} 0, & d = 0, \\ k_{\mathrm{mat}} d, & 0 < d < d_c, \\ F^{\mathrm{cut}}_{\mathrm{mat}}, & d \geq d_c. \end{cases} \tag{66}$$

The first branch represents separation, the linear branch represents material loading before fracture, and the plateau represents continued cutting after the force threshold has been reached.

Proposition 7 (Boundedness and regularity of the identified cutting law)**.** *For $k_{\mathrm{mat}} > 0$ and $F^{\mathrm{cut}}_{\mathrm{mat}} > 0$, the scalar law* (66) *is non-negative, monotone non-decreasing, globally bounded by $F^{\mathrm{cut}}_{\mathrm{mat}}$ and globally Lipschitz on $\mathbb{R}_{\geq 0}$ with constant k_{mat}. Its scalar work primitive is*

$$U_{\mathrm{mat}}(d) = \int_0^d f_{\mathrm{mat}}(s)\,ds = \begin{cases} \frac{1}{2} k_{\mathrm{mat}} d^2, & 0 \leq d \leq d_c, \\ F^{\mathrm{cut}}_{\mathrm{mat}} d - \dfrac{(F^{\mathrm{cut}}_{\mathrm{mat}})^2}{2k_{\mathrm{mat}}}, & d > d_c. \end{cases} \tag{67}$$

Proof. Both arguments of the minimum in (65) are non-negative and non-decreasing for $d \geq 0$, which establishes non-negativity and monotonicity. The minimum is at most $F^{\mathrm{cut}}_{\mathrm{mat}}$. For any $d_1, d_2 \geq 0$,

$$|f_{\mathrm{mat}}(d_1) - f_{\mathrm{mat}}(d_2)| \leq k_{\mathrm{mat}} |d_1 - d_2|, \tag{68}$$

because saturation cannot increase the slope of the unsaturated linear map. Integration of the two branches gives (67); at $d = d_c$, both expressions equal $(F^{\mathrm{cut}}_{\mathrm{mat}})^2/(2k_{\mathrm{mat}})$, proving continuity. □

Parameter identification. For each material, let $\mathcal{D}_{\mathrm{mat}} = \{(d_j, y_j, \omega_j)\}_{j=1}^{N_{\mathrm{mat}}}$ contain the penetration d_j, measured force magnitude $y_j = \|F_j\|$ and non-negative sample weight ω_j. The two material parameters are obtained from the constrained nonlinear least-squares problem

$$\hat{\theta}_{\mathrm{mat}} = \arg \min_{k>0,\, F^{\mathrm{cut}}>0} \sum_{j=1}^{N_{\mathrm{mat}}} \omega_j \left[y_j - \min\big(k d_j, F^{\mathrm{cut}}\big)\right]^2. \tag{69}$$

For a fixed division into a loading set $\mathcal{L} = \{j : k d_j < F^{\mathrm{cut}}\}$ and plateau set $\mathcal{P} = \{j : k d_j \geq F^{\mathrm{cut}}\}$, the estimates have the closed forms

$$\hat{k}_{\mathrm{mat}} = \frac{\sum_{j\in\mathcal{L}} \omega_j d_j y_j}{\sum_{j\in\mathcal{L}} \omega_j d_j^2}, \qquad \hat{F}^{\mathrm{cut}}_{\mathrm{mat}} = \frac{\sum_{j\in\mathcal{P}} \omega_j y_j}{\sum_{j\in\mathcal{P}} \omega_j}. \tag{70}$$

These expressions show separately how the loading samples determine stiffness and how the steady cutting samples determine the plateau. The partition is updated using $d_c = \hat{F}^{\mathrm{cut}}_{\mathrm{mat}}/\hat{k}_{\mathrm{mat}}$ until the branch assignments are consistent. A unique estimate for a fixed non-empty partition requires $\sum_{j\in\mathcal{L}} \omega_j d_j^2 > 0$ and $\sum_{j\in\mathcal{P}} \omega_j > 0$.

The demonstrated reaction direction can be estimated independently from the normalised force samples:

$$\hat{F}_d(\phi) = \frac{\sum_{j\in\mathcal{N}(\phi)} \omega_j\, F_j/(\|F_j\| + \varepsilon_F)}{\left\|\sum_{j\in\mathcal{N}(\phi)} \omega_j\, F_j/(\|F_j\| + \varepsilon_F)\right\| + \varepsilon_r}, \tag{71}$$

where $\mathcal{N}(\phi)$ is a neighbourhood of the phase and $\varepsilon_F, \varepsilon_r > 0$ regularise small measured forces and a near-zero directional resultant. This separates identification of the scalar material strength from identification of the demonstrated force direction; the reaction direction applied to the blade is $-\hat{F}_d(\phi)$ as in (65).

The geometric box contact is disabled during the cutting experiment, F_{react} from (65) is applied to the blade, and the force sensor reports the same interaction. Parameter sensitivity is bounded by

$$|f_{\mathrm{mat}}(d; k_1, F_1) - f_{\mathrm{mat}}(d; k_2, F_2)| \le d|k_1 - k_2| + |F_1 - F_2|, \tag{72}$$

so errors in the identified loading stiffness dominate at larger pre-fracture penetrations, whereas plateau error directly bounds the post-fracture force error.

Material	$F^{\mathrm{cut}}_{\mathrm{mat}}$ (N)	k_{mat} (N/m)	contact maintained	force RMSE (N)
penoplex (foam)	63	9,104	100%	1.8
cork	59	9,917	100%	2.5
PVC (hardest)	86	13,855	100%	2.6

Table 5: Cutting parameters identified per material, and the resulting force tracking. The materials are clearly distinct (PVC is stiffest and strongest, consistent with its hardness), and the force matches the demonstrated level to within ≈ 2 N — an approximate physical fit, not an exact replay.

7 Results

Figure 7 summarises the closed-loop behaviour with the identified material on both a static and a moving workpiece.

The same behaviour holds for the other two materials (Fig. 8): on **PVC** the force tracks its higher $\sim$86 N cutting plateau and on **penoplex** its $\sim$63 N plateau, each within ≈ 2 N RMSE and 100% contact, on both static and moving workpieces. The force level is thus material-specific (set by the identified $F^{\mathrm{cut}}_{\mathrm{mat}}$ of Table 5) while the same controller and the same identification procedure are used throughout.

- **The task performance is attributable to the learned variance structure** (Table 3, Fig. 3): the identified precision attains the lowest cut-depth error — 26–48% below the inverted, permuted, and direct-force assignments — on every stable seed ($p = 0.001$).
- **Force tracking** (Table 5, Fig. 7): with the identified material the measured force follows the demonstrated force to ≈ 2 N RMSE with 100% contact, on both static and moving workpieces and across three materials.
- **Position tracking**: cut-depth (Z) error ≈ 2 mm; the lateral (Y) error peaks ≈ 50 mm during the large ± 150 mm demonstrated lateral swing — the expected consequence of the deliberately low stiffness assigned to that low-precision axis.

- **Passivity** (Fig. 4, Table 4): under positive joint-power perturbation, the controller without tank projection violates the available energy budget for 12.0% of the samples (+5876 W maximum margin), while both tanked controllers satisfy $\dot{q}^\top \tau_{\text{safe}} \le P_{\max}$ for the whole run.
- **Stability diagnostics** (Fig. 5): nominal static and moving cuts remain bounded and complete the task; cut-depth error stays small. The exponential fit is reported as a diagnostic, while the full Cartesian error norm is interpreted carefully because low-precision axes are intentionally released.
- **Run monitoring**: Figure 9 shows the monitored signals used to inspect runs: position, contact force, stiffness, tracking error and tank energy, each in its own units.

8 Novelty and contributions

1. **Gains learned from demonstration precision, via an online QP.** Stiffness *is* the demonstration precision (20), computed each step by the QP (21) that blends force-consistency with precision-stiffness under force-saturation constraints. The controlled comparison (Table 3) shows the *specific* learned variance structure — not generic adaptivity — is what carries the skill.
2. **State-driven task parametrisation.** A phase variable advanced by matching the robot's measured state to the prior (42), rather than by a clock, giving speed-adaptability and disturbance awareness, and unifying with the stability mechanism.
3. **Torque-port passivity under variable gains and a moving target.** The live energy tank sets the torque-level power budget (45); Fig. 4 demonstrates that this rejects positive-power perturbations. A cross-term Lyapunov proof (Theorem 1) gives the exponential-contraction condition for the regulated error dynamics, and the tank provides the mechanism that bounds phase and gain rates.
4. **An empirical principle for contact-skill transfer.** Position is the reproducible prior, force is not; the controller therefore regulates position and treats force as a constraint, while a per-material identified model supplies the interaction force.

9 Assumptions, scope and limitations

- The contact *force* matches because the material model is identified to the demonstration (system identification); the *controller* achieves the position/velocity tracking. These two roles are reported separately, and the force result is *not* presented as controller force-tracking.
- All experiments are in simulation; the material parameters are identified from the real material's measured demonstration forces. Real-robot validation is future work.
- Lateral (Y) tracking lags the large demonstrated lateral swing; the task-critical cut-depth is tight. Reducing the lateral error is a phase-speed / parametrisation question, separate from the force model.
- The energy-tank passivity guarantee is implemented and validated at the joint torque port. The reconstructed Cartesian residual is retained as a useful diagnostic, but it is not the primary proof artifact until the force frame/sign, finite-difference velocity estimate and Cartesian storage model are matched more precisely.

- The exponential-stability proof applies under regulation assumptions (constant or sufficiently slowly varying gains, passive environment, bounded reference acceleration). The current simulations support bounded stable cutting; a stronger empirical exponential-convergence claim should be made on the regulated cut-depth or precision-weighted error, not on the full Cartesian norm that includes intentionally released axes.

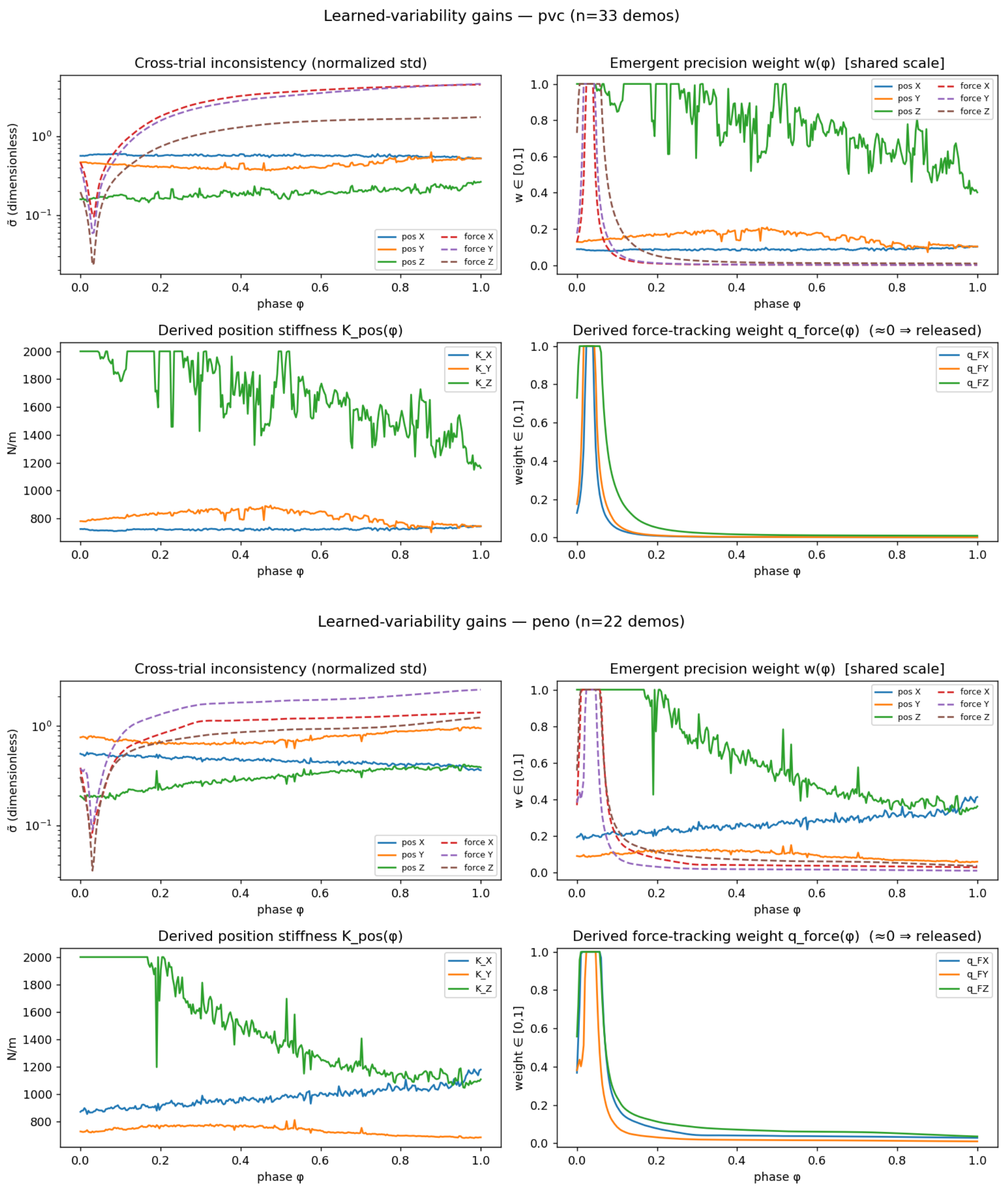

Figure 2: Learned gains for PVC (top) and penoplex (bottom), produced by the same precision rule (20) on each material's own demonstrations. The structure is consistent with cork (Fig. 1) — the depth axis is tracked stiffly and force is released — while the stiffness magnitudes and profiles differ per material, so the controller expresses material-specific behaviour without any manual retuning.

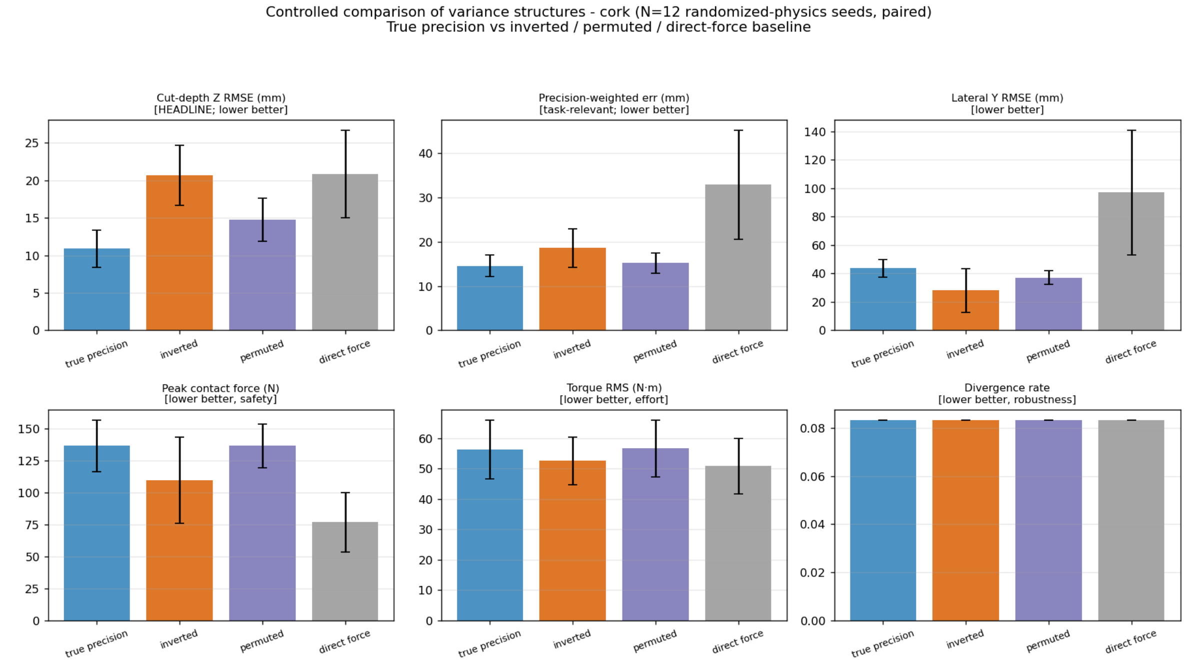

Figure 3: Controlled comparison of gain assignments on cork ($N_{\text{seed}} = 12$ randomised-physics seeds, paired). Only the identified precision attains low cut-depth error; inverting or permuting the learned structure degrades it. Bars denote mean±std.

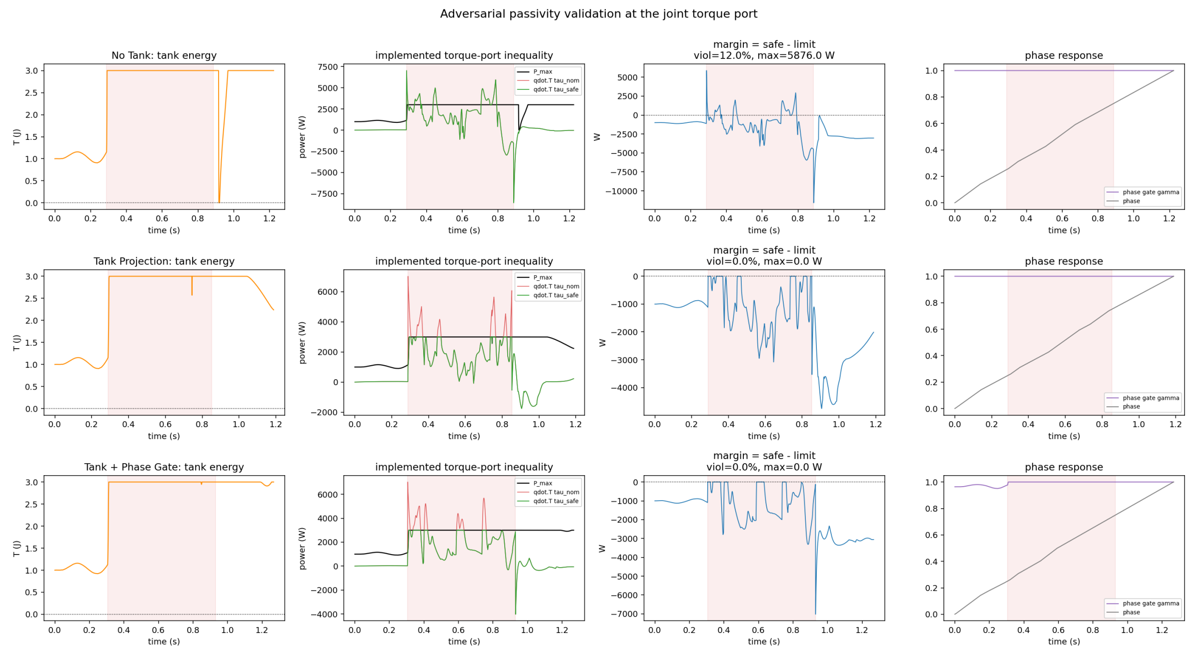

Figure 4: Positive-power perturbation validation on cork. A known positive joint-power burst is injected before the torque projection. Without the tank (top row), $\dot{q}^{\top}\tau$ exceeds the counterfactual budget for 12.0% of the samples and reaches a maximum margin of +5876 W. With the tank projection (middle) and with tank plus phase gating (bottom), the implemented inequality $\dot{q}^{\top}\tau_{\text{safe}} - P_{\max} \leq 0$ is satisfied for the whole run (0.0% violations). The red shading marks the perturbation window.

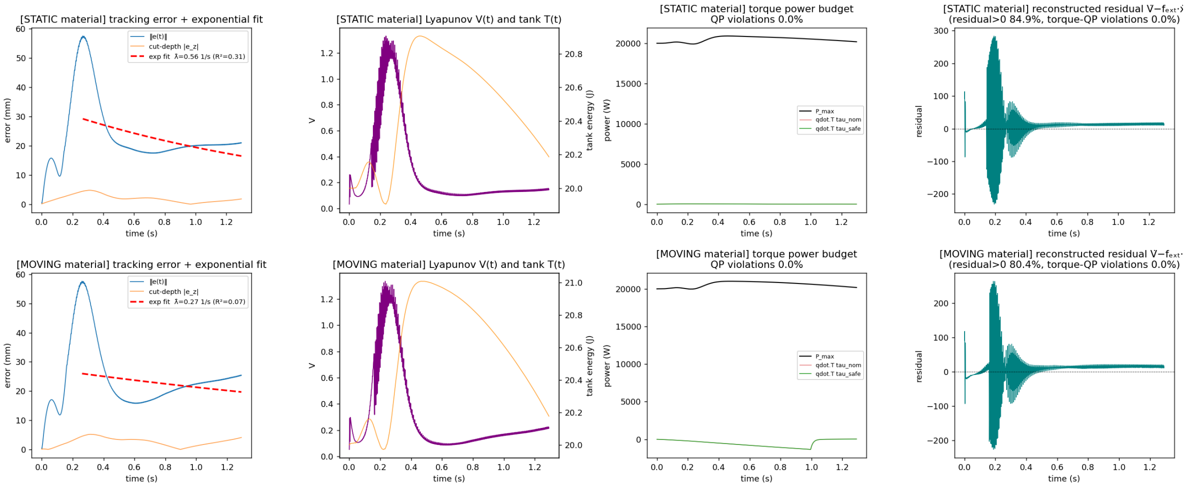

Figure 5: Nominal stability and passivity diagnostics on cork. The left column shows tracking error and cut-depth error; the middle column shows the Lyapunov diagnostic and tank energy; the third column shows the implemented torque power budget $P_{\max}$ together with nominal and projected torque power; the right column shows the reconstructed Cartesian residual, used as a diagnostic rather than as the primary passivity proof.

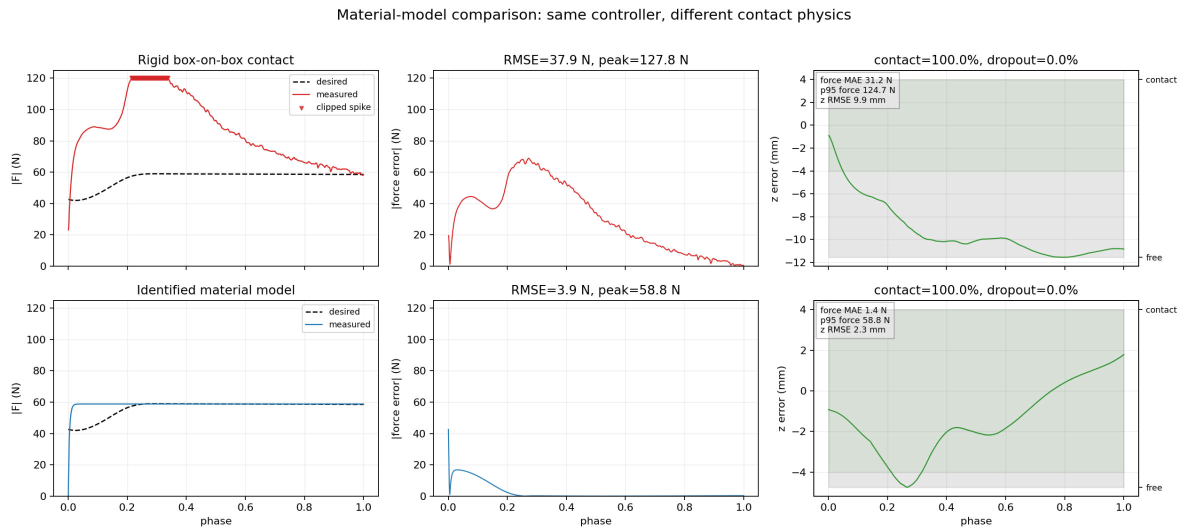

Figure 6: Why a material model is needed. This matched cork comparison uses the same controller and toggles only the contact physics. Top row: rigid box-on-box contact over-reacts and follows the solver geometry rather than the demonstrated cutting plateau (37.9 N force RMSE, 127.8 N peak, 9.9 mm depth RMSE). Bottom row: the identified cutting-force law produces the bounded material plateau and tracks the demonstrated force (3.9 N force RMSE, 58.8 N peak, 2.3 mm depth RMSE).

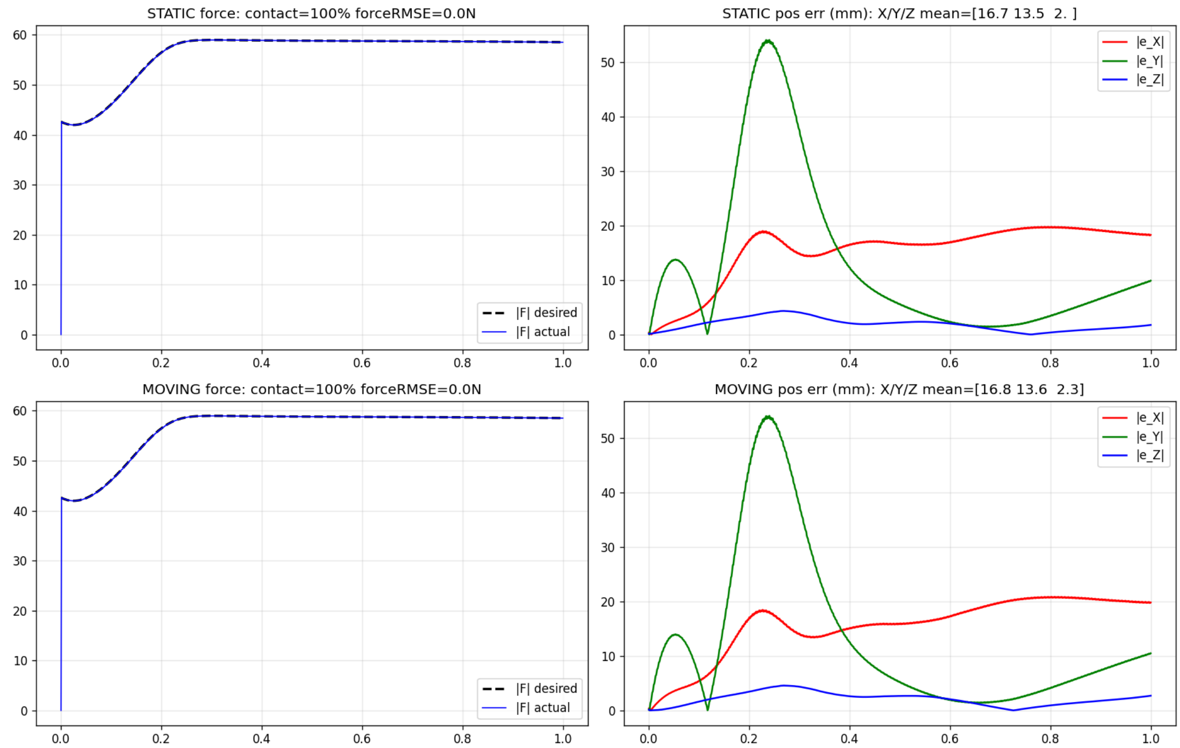

Figure 7: Cutting with the identified material model on **cork**. Left column: the measured contact-force magnitude (blue) follows the desired (dashed) for the whole cut, on static (top) and moving (bottom) workpieces. Right column: per-axis position error — the cut-depth (Z) error stays ≈ 2 mm; the lateral (Y) error peaks during the large demonstrated lateral swing.

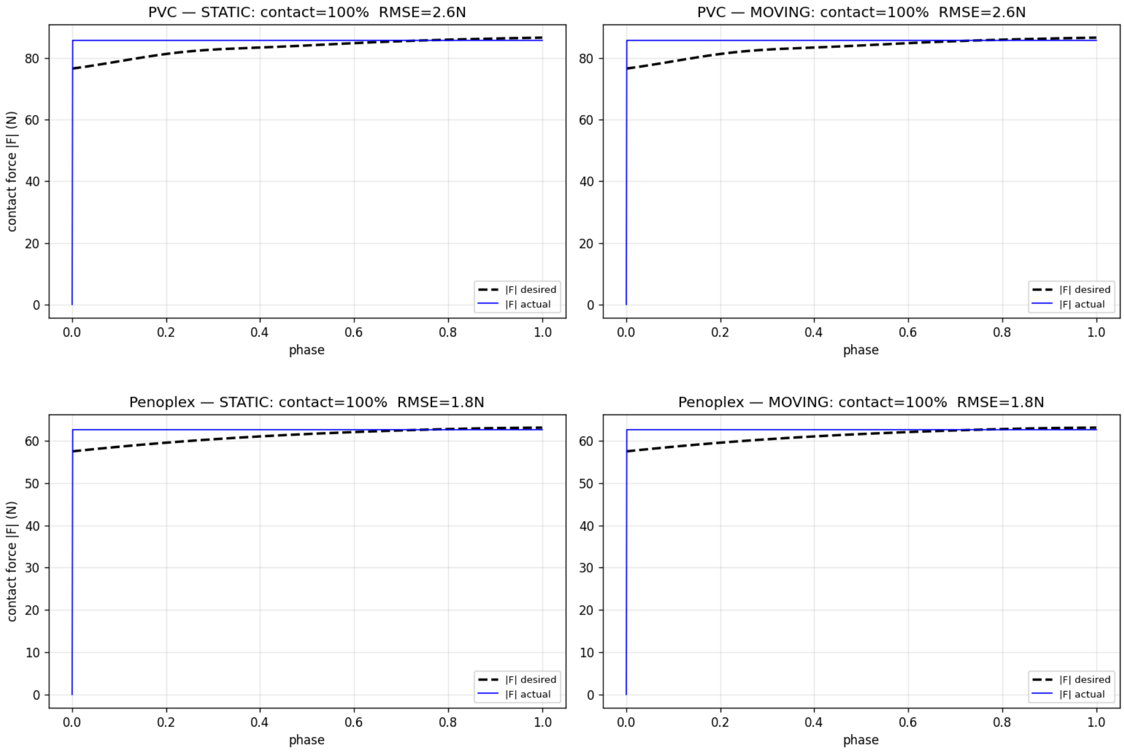

Figure 8: Force tracking on **PVC** (top) and **penoplex** (bottom), static (left) and moving (right). The measured force (blue) follows each material's demonstrated level (dashed) to $\approx 2\,\mathrm{N}$ RMSE with 100% contact. PVC plateaus highest ($\sim 86\,\mathrm{N}$, hardest material), penoplex lowest ($\sim 63\,\mathrm{N}$), mirroring the per-material cutting parameters of Table 5.

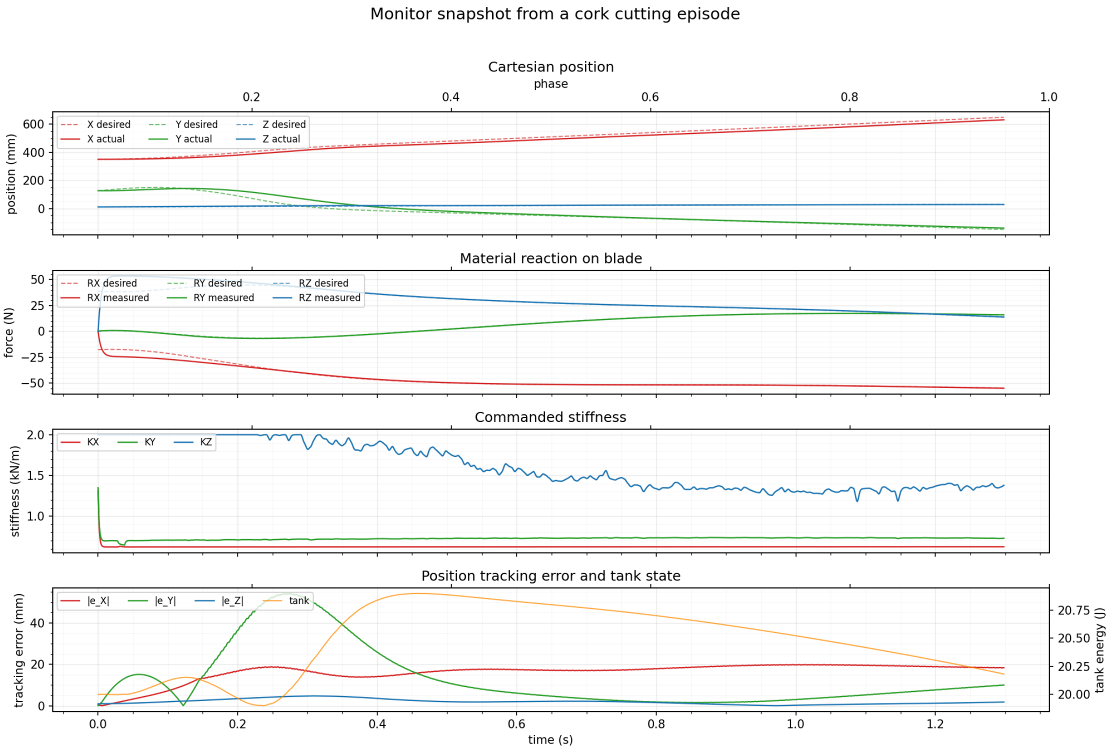

Figure 9: Monitor snapshot reconstructed from one cork cutting episode (1299 controller samples, final phase 1.000). The panels show Cartesian position (mm), material reaction force per axis (N, desired vs measured), commanded stiffness (kN/m), and position tracking error (mm) together with tank energy (J).